

Math 420: Final Project

December 5th, 2025

For the in class project, pick one of the three options below to complete. The project is due December 5th at Midnight. This project will allow us to apply the concepts from class to help motivate why they might be helpful when pursuing future careers, hobbies, or mathematics. Have fun with your project and be as creative as you wish!

1 Lesson plan

One of the best ways to learn a concept is to teach it. For this project, I would like you to make a lesson plan for a high school or beyond level. Your lesson plan should explain the concept of your choice using ideas from linear algebra (even if you would not use the formal language of linear algebra explicitly when teaching your concept). In addition to your lesson plan, submit a write up explaining which concepts from linear algebra were used in your lesson. In your write-up, be sure to prove any claims made in your lesson. If you are a math education major, you may have specific formatting given in previous classes. Feel free to use these types of formatting as long as you include all the necessary concepts. The goal of this option is to give practice planning lessons before you teach in a classroom.

Things to turn in:

1. The lesson plan. The format of this is very flexible, but it should be a lesson for at least 1 hour of content on a linear algebra topic.
2. A write-up report of at least 2 pages. This write-up should summarize the lesson plan, give details and explanation on the linear algebra concepts used, and prove any claims made in the lesson.

2 Deep dive into application

Linear algebra has many applications outside of pure mathematics. Pick an application we did not discuss in class, and explain the application in detail. In addition, prove any claims made in your project or provide citations of proofs. (Some applications have very complicated proofs. If you have any questions about what level of rigor is appropriate, please ask me over email or during my office hours).

Thing to turn in:

1. A write-up report of at least 4 pages. This write-up should explain in detail the application, give and describe the linear algebra concepts used, provide examples of this connection, and prove any claims made about linear algebra.

3 Explore a topic

Linear algebra has many topics that can be explored further. For this option, pick a linear algebra topic and explore it in detail. This can be done either in the form of a technical paper or as a presentation style format. Similar to the application option, proofs/concepts can get very difficult depending on the direction. Please reach out if you are unsure on the level of rigor required.

Thing to turn in:

1. A write-up report of at least 4 pages or a presentation for a 1 hour+ lecture. This write-up should explain in detail the topic, give and describe linear algebra concepts used, provide examples of this topic, and prove any claims made.

4 Historical/Current usage

Linear algebra has a long and interesting history. It has been used to solve many difficult problems throughout history and is still an active field of research today. Pick a historical or current problem or theorem in math that is related to linear algebra. Explain your problem/theorem in appropriate terms. If the problem has been solved, explain the proof and what steps were taken in the process of solving the problem. If the problem is currently unsolved, write about some of the attempts made to solve the problem, current research into the problem, and how the problem came about. If you chose a theorem, be sure to provide a proof or a sketch of a proof of your theorem. Your explanation should use the appropriate terminology from class and should define any concepts not discussed in class.

Thing to turn in:

1. A write-up report of at least 4 pages. This write-up should explain in detail the historical usage of linear algebra on some problem or field, provide a sketch of the proof if the problem has been solved or a sketched attempted proof if it is open, and give some details on the ramifications of the problem.